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Lecture Notes on Differential Equations and Transform Techniques

Spring 2025

The purpose of these lecture notes is to provide a concise summary of the key concepts covered in the course. They are designed to serve as a **quick reference** and should be used in conjunction with the core textbooks listed below. While some sections are adapted from these referenced texts, the notes are not intended to replace the original works, which offer more comprehensive explanations. To gain a deeper understanding, students are encouraged to refer to the textbooks listed.

References:

1. W. E. Boyce, R. C. DiPrima, D. B. Meade, *Elementary Differential Equations and Boundary Value Problems*, 9th edition, Wiley, 1997.

2. Brad G. Osgood, *Lectures on Fourier Transform and Its Applications*, IAS/Park City Mathematics Series / American Mathematical Society (AMS), 2019.

3. Charles Henry Edwards, David E. Penney, *Differential Equations and Boundary Value Problems: Computing and Modeling*, 4th edition, Pearson, 2000.

4. George F. Simmons, *Differential Equations with Applications and Historical Notes*, 2nd edition, CRC Press, 2016.

5. Ruel V. Churchill, *Fourier Series and Boundary Value Problems*, McGraw-Hill, 1941.

6. Gustav Doetsch, *Introduction to the Theory and Application of the Laplace Transformation*, Springer, 2012.

7. Morris Tenenbaum, Harry Pollard, *Ordinary Differential Equations*, Dover Books on Mathematics, Dover Publications, 1985.

EE1060: Differential Equations and Transform Techniques

Lecture Notes - 23

Topics covered :

1. Introduction to Fourier Transform [1]

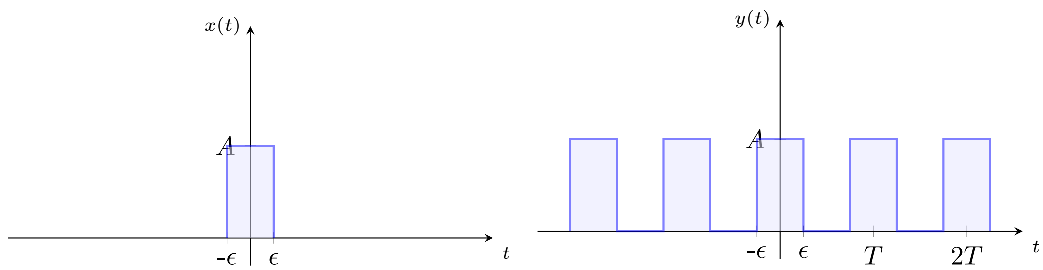
2. Inverse Fourier Transform

- The Fourier series expansion expresses a periodic function as a sum of complex exponentials. However, for non-periodic signals, this complex exponential decomposition cannot be obtained through Fourier series.

- One method to determine the complex exponentials that synthesize a non-periodic signal is to recognize that a non-periodic signal can be represented by first constructing a periodic extension and then taking the limit as the period approaches infinity (∞). If $x(t)$ represents a non-periodic signal, a periodically extended signal $y(t)$ with period T can be generated from the original signal as

$$y(t) \;=\; \sum_{k=-\infty}^{\infty} x\left(t + kT\right) \tag{1}$$

For instance, consider a basic rectangular pulse with width ϵ, as shown in Fig. 1(a). The periodic extension of this signal, generated using Eq. 1, results in a signal $y(t)$ as shown in Fig. 1(b). It can evident that as $T \to \infty$, the



(a) Non-periodic signal

(b) Periodic extension

Fig. 1: Non-periodic signal and its extended periodic signal

periodic extension of the signal converges to the original signal $x(t)$.

- To begin, let us examine the Fourier series of the signal $y(t)$ and analyze its behavior as T increases. The Fourier

[1] Fourier Transforms will be explored in detail in the Signals and Systems course. The aim here is to introduce the fundamental definition and familiarize you with the idea that the Fourier Transform can be viewed as an extension of the Fourier Series.

coefficient of the signal $y(t)$ can be computed as

$$\begin{aligned}
C_k &= \frac{1}{T}\int_{-\epsilon}^{\epsilon} A e^{-jk\omega_0 t}\ dt \\
&= \frac{A}{T(-jk\omega_0)}\ e^{-jk\omega_0 t}\Big|_{-\epsilon}^{\epsilon} \\
&= \frac{jA}{2k\pi}\left[e^{-jk\epsilon\omega_0} - e^{jk\epsilon\omega_0}\right] \\
&= A\frac{\sin\left(k\epsilon\omega_0\right)}{k\pi} \;=\; \frac{2A}{T}\frac{\sin(\omega\epsilon)}{\omega} \text{ where } \omega = k\omega_0
\end{aligned} \tag{2}$$

The magnitude spectrum of the signal $y(t)$ for various values of T is shown in Fig. 2. While the magnitude spectrum is typically plotted for different values of k, we express the Fourier coefficients and plot the spectrum as a function of frequency (i.e., $k\omega_0$). From the frequency spectrum shown in Fig. 2, the following observations can

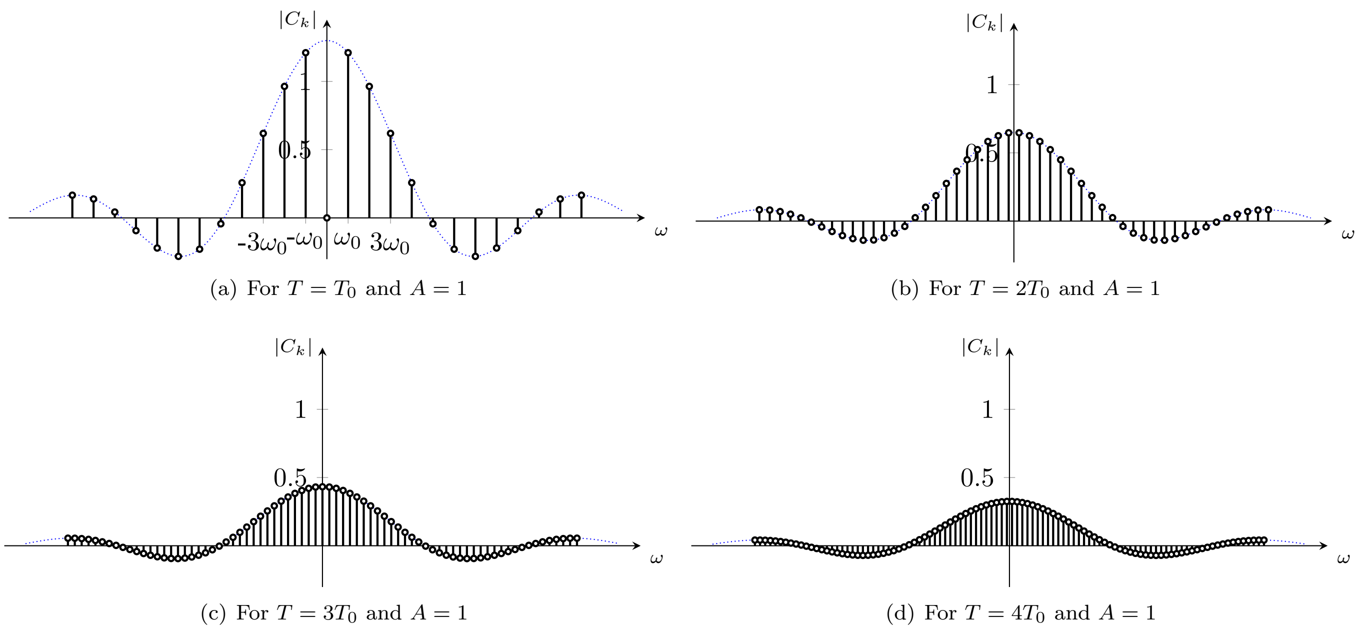


Fig. 2: Magnitude of Fourier coefficients

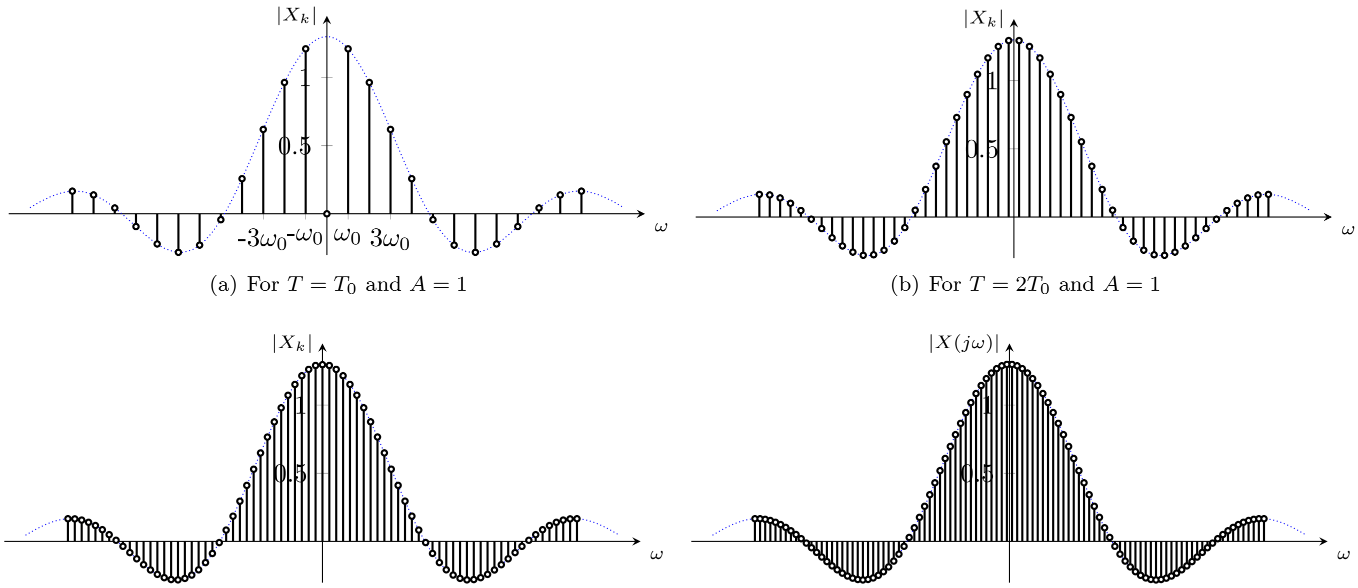
be made:

- As the time period T increases, the fundamental frequency ω_0 reduces and the spacing between the frequency bins in the magnitude spectrum narrows. In other words, for a given frequency range, the number of bins increases as the time period increases. It can be inferred that as $T \to \infty$, the bins become increasingly close, resulting in a continuous frequency spectrum.
- The magnitude spectrum for different frequencies appears to be a sampled version of a continuous signal. However, as T increases, the magnitude of the coefficients tends to decrease. This suggests the need to reconsider the interpretation of the Fourier coefficients.

• Instead of considering C_k, we now examine the product $C_kT = X(j\omega)$ as an indicator of the harmonic spectrum. In this context, it can be observed (see Fig. 3) that the harmonic components resemble the result of sampling a continuous-time signal as a function of ω. As $T \to \infty$, the harmonic components progressively approximate the continuous-time signal. Therefore, the value of $X(j\omega)$ as $T \to \infty$ can be written as

$$\begin{aligned} X(j\omega) &= \lim_{T\to\infty} TC_k = \lim_{T\to\infty} \int_{-\frac{T}{2}}^{\frac{T}{2}} x(t)e^{-jk\omega_0 t}\, dt \\ &= \int_{-\infty}^{\infty} x(t)e^{-j\omega t}\, dt \end{aligned} \tag{3}$$

Thus, $X(j\omega)$ can be interpreted as the frequency spectrum of the non-periodic signal $x(t)$, leading to the concept of the Fourier Transform.



(a) For $T = T_0$ and $A = 1$

(b) For $T = 2T_0$ and $A = 1$

(c) For $T = 3T_0$ and $A = 1$

(d) For $T = 4T_0$ and $A = 1$

Fig. 3: Magnitude of Fourier coefficients TC_k

- The Fourier transform of a non-periodic signal $x(t)$ is defined as

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}\, dt \tag{4}$$

- The inverse Fourier Transform (or equivalently, the synthesis equation) is used to reconstruct a signal from its Fourier Transform coefficients $X(j\omega)$. To better understand this, we begin with the Fourier series synthesis equation, which expresses a periodic signal as a sum of complex exponentials

$$x(t) = \sum_{k=-\infty}^{\infty} \frac{1}{T} X(j\omega)e^{jk\omega_0 t} = \sum_{k=-\infty}^{\infty} \frac{\omega_0}{2\pi} X(j\omega)e^{j\omega t} \tag{5}$$

However, as we have observed earlier, when the period $T \to \infty$, the fundamental frequency $\omega_0$becomes negligibly small, and the summation turns into an integral. Thus, in the limit as $T \to \infty$, the summation in the Fourier series turns into an integral, and we can write the signal $x(t)$ as

$$x(t) \;=\; \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)e^{j\omega t}\,d\omega \tag{6}$$

Equation (6) represents the Inverse Fourier Transform, which reconstructs the original signal from its Fourier Transform coefficients.